Modeling and Taguchi based analysis of Innopreneurship Process, A Multi-Concept Perspective for Sustainable Education

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Abstract

Innopreneurship is the practice of innovation and entrepreneurship is becoming increasingly important in today's dynamic world of education. Universities are ideal places for innopreneurship because they offer a wealth of resources, including talented students, knowledgeable faculty and advanced research facilities. By analyzing the relationship between these variables, leads to the success of innopreneurship in universities, it is possible to identify which factors are most important in promoting successful innovation-driven entrepreneurship is the need of the education system. This paper has made the effort in modelling of innopreneurship with different variables using Taguchi method in higher education institutes. The outcome of this information can be used to develop strategies and policies to enhance the university's role in promoting innopreneurship. In total promoting innopreneurship in universities can help to create a pipeline of innovative startups and entrepreneurs who can drive economic growth and create new jobs. Based on the Taguchi analysis the variable Infrastructure and facilities (X2) has highest influence and Promotion and Supporting Entrepreneurship (X4) has lowest influence than other parameters on Innopreneurship process.

Keywords: Innovation, Entrepreneurship, Innopreneurship, Taguchi Method.

Introduction

Innopreneurship, also known as innovation-driven entrepreneurship, refers to the process of combining innovation and entrepreneurship to create new products, services or business models. It involves identifying new opportunities, developing innovative solutions, and bringing them to market. In the context of universities, innopreneurship refers to the creation of a culture and ecosystem that fosters innovation and entrepreneurship among students, faculty, and staff. Here are some ways universities can promote innopreneurship:

I. Entrepreneurship programs: Walter, S.G. (2016), Universities can offer entrepreneurship programs that provide students with the necessary knowledge, skills, and resources to start their own businesses. These programs can include courses, workshops, mentoring, and networking opportunities.
II. Incubators and accelerators: Universities can establish incubators and accelerators that provide startups with support services, such as office space, access to funding, mentorship, and networking opportunities.
III. Research and development: Universities can promote innopreneurship by encouraging research and development activities that lead to new products, services, or business

models. Universities can also collaborate with industry partners to commercialize research findings.

IV. Entrepreneurial culture and Funding: Universities can create a culture of innovation and entrepreneurship by encouraging students and faculty to take risks, think outside the box, and pursue their entrepreneurial ideas. This can be achieved through events, competitions, and other activities that promote entrepreneurship.

V. Partnerships with industry: Universities can partner with industry to provide students with opportunities to work on real-world problems and develop innovative solutions. These partnerships can also provide students with access to industry mentors and potential funding sources.

VI. Encourage innovation: Universities can encourage students, faculty, and staff to think creatively and generate new ideas by providing opportunities for innovation. This can include hosting innovation challenges, hackathons, and ideation sessions.

VII. Intellectual property protection: This variable measures the extent to which universities provide support for the protection and commercialization of intellectual property.

Yuanyuan Yang (2024), Promoting innopreneurship in universities can help to create a pipeline of innovative startups and entrepreneurs who can drive economic growth and create new jobs. Figure-1 will explain about Innopreneurship which brings the novelty in entrepreneurial activities that create the value. Innovation enables the entrepreneurs to deal with new technology, and new ways to take strategic decisions. Innopreneurship is the entrepreneurial attitude that balances the risk in university education with novel skills and ideas.

Literature

André Cherubini Alves, Bruno Fischer (2018) Current research paper was conducted with the objective of identifying the traits and characteristics of student entrepreneurship across various universities and departments. Research paper suggested a model consisting of different dimensions like entrepreneurial activity, potential entrepreneurs, high-impact entrepreneurship, serial entrepreneurship and innovation-driven entrepreneurship etc.

Alexandor Harsono (2013) Importance of Technopreneurship education in building students career has been explored in the current research paper. Researcher mentioned that entrepreneurship is a discipline and it can be learned and it has nothing to do with genes.

Xingjian Wei,, Liu, and Jian Sha (2019) This study was oriented towards to understand the mediating effect of entrepreneurship education on entrepreneurial innovation. Study shows that perception of entrepreneurial education has mediating effect on perception of innovation.

Kolawole Shola Ojo (2023) Role of self-monitoring and resilience in innovation was studied through current research article. Author examined the model of componential theory of creativity. It was observed that this theory suggesting two important aspects that determines the innovation among the students, those are individual resilience and self-assessment.

Faisal Iddris (2016) Researcher conducted a literature review-based study to examine the innovation capacity building initiatives taken by the business organizations. Researcher studied several research papers and concluded that there are various dimensions of innovation capability building. Some of the factors identified through the review includes

vision and mission of the organization, knowledge generation capability, Knowledge retaining capability etc.

Wang and Chugh (2014) demonstrated in their systematic literature review, previous studies on entrepreneurial learning (EL) points to a variety of theoretical constructs and research perspectives, including experiential and organizational learning. Different ways of learning can occur, each on a binary continuum reaching from individual to collective learning, exploratory to exploitative learning, intuitive to sensing learning

Innopreneurship Spirit

Innopreneurship blends innovation and entrepreneurship, fostering a mindset geared toward developing ground breaking outputs, services, or processes that drive sustainable multi-concept education businesses. Universities play a vital role in cultivating the innopreneurial spirit among students, faculty, and staff by providing the necessary resources, support, and opportunities for innovation-driven ventures. Figure 01 shows one such process for the study.

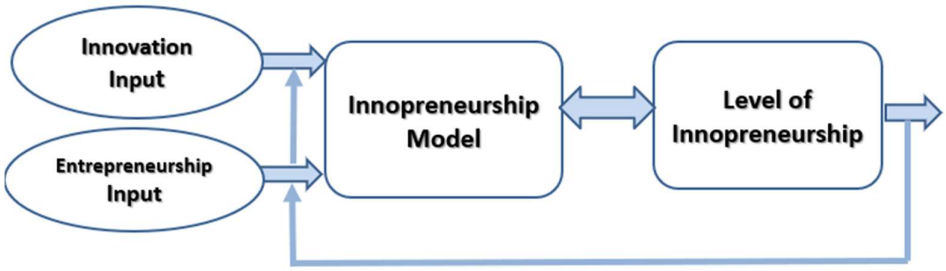


Figure 01. Innopreneurship frame work model.

Promoting an innopreneurial spirit in universities can help develop the next generation of innovative leaders and entrepreneurs who can drive economic growth and social change. Benchmarking the Innopreneurship spirit of a university would involve assessing the university's ability to foster innovation, creativity, and entrepreneurial thinking among its students, faculty, and staff.

Objectives:

a) To identify the key strategies employed by private universities to foster entrepreneurial spirit and Innopreneurship among students.
b) To study student's inclination towards the innovation and entrepreneurship.
c) To provide practical recommendations for educators, policymakers, and university administrators about Innopreneurship spirit at any point of the process.

Regression Modelling of Innopreneurship

Innopreneurship, or innovation-driven entrepreneurship, refers to the process of creating new businesses or ventures that are driven by innovative ideas or technologies. Universities can play a significant role in promoting innopreneurship by fostering an environment that encourages creativity, collaboration, and experimentation. One way to model innopreneurship in universities is through a regression analysis that examines the relationship between different variables and the likelihood of successful innovation-driven ventures Sadan Kulturel(2024).

By analyzing the relationship between different variables will lead to the success of innopreneurship in universities, it is possible to identify which factors are most important in promoting successful innovation-driven ventures. This information can be used to develop strategies and policies that can further enhance the university's role in promoting innopreneurship.

A Regression model for the Innopreneurship can be represented as a formula or a set of equations that describe the various factors that contribute to the level of innovation in a country, region, or company. The model typically considers factors such as research and development expenditure, patent filings, human capital, infrastructure, and other factors that impact innovation. For instance, one regression model for the Innopreneurship is formulated as shown.

Regression model

Innopreneurship =3.38 + 0.208 X1 - 0.982 X2 - 0.125 X3 + 0.075 X4 + 0.218 X5 + 0.218 X6 + 0.172 X7

Table 01: Coded Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	3.057	0.142	21.57	0.000	
X1	0.104	0.142	0.73	0.503	1.00
X2	-0.491	0.142	-3.46	0.026	1.00
X3	-0.063	0.142	-0.44	0.682	1.00
X4	0.037	0.142	0.26	0.804	1.00
X5	0.109	0.142	0.77	0.484	1.00
X6	0.109	0.142	0.77	0.484	1.00
X7	0.086	0.142	0.61	0.578	1.00

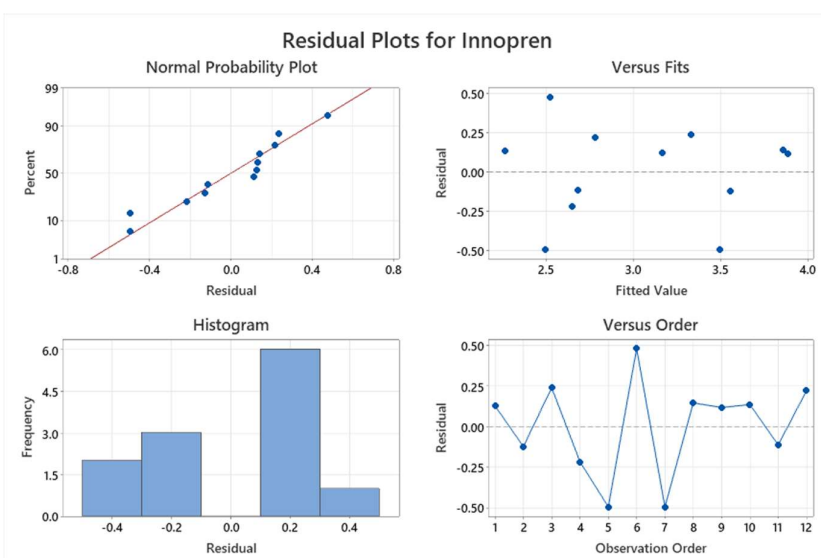
Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.491079	78.20%	40.04%	0.00%

Table 02: Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	7	3.45939	0.49420	2.05	0.255
X1	1	0.13021	0.13021	0.54	0.503
X2	1	2.89101	2.89101	11.99	0.026
X3	1	0.04688	0.04688	0.19	0.682
X4	1	0.01687	0.01687	0.07	0.804
X5	1	0.14301	0.14301	0.59	0.484
X6	1	0.14301	0.14301	0.59	0.484
X7	1	0.08841	0.08841	0.37	0.578
Error	4	0.96463	0.24116		
Total	11	4.42403			

Figure-02 Residual plots



This model will be used to rank and compare the innovation performance of different entities and identify areas of improvement to enhance innovation and Entrepreneurship Capabilities. However, the model's effectiveness depends on the accuracy and relevance of the input data and the appropriateness of the weights assigned to different factors for the study.

Taguchi Method

Taguchi method is a statistical technique used to improve the quality of a product or process by identifying the key factors that affect its performance and optimizing them. In the context of innopreneurship, the Taguchi method can be applied to the innopreneurship process of a university to enhance the quality and effectiveness of its teaching learning and student outcome or products or services. Here are some ways in which the Taguchi method can be applied in the innopreneurship process of a university:

I. Identify key factors: The first step in applying the Taguchi method is to identify the key factors that affect the quality of the product or service. In the context of a university, this could include factors such as the quality of the research, the skills of the researchers, the availability of resources, and the effectiveness of the collaboration between researchers and industry partners.
II. Design experiments: The next step is to design experiments to test the impact of each factor on the outcome of the innovation process. This can be done using statistical techniques to ensure that the experiments are robust and produce reliable results.
III. Conduct experiments: The experiments should be conducted according to the design, and the results should be recorded and analyzed. This will help to identify the factors that have the greatest impact on the quality of the innovation process.
IV. Optimize the process: Once the key factors have been identified, the Taguchi method can be used to optimize the innovation process by adjusting the levels of each factor to achieve the best possible outcome. This can involve changing the research methodology, providing additional resources, or improving the collaboration with industry partners.

Taguchi method is a valuable approach for enhancing the quality and efficiency of the innopreneurship process in universities. By pinpointing and optimizing key factors influencing innovation and entrepreneurship outcomes, universities can strengthen their capacity to cultivate innovative students and initiatives that effectively address stakeholder needs. Based on the survey following key factors considered for Innopreneurship study are list below:

I. Budget, Expenses to Support & Revenue Generated [X1].
II. Infrastructures & Facilities to Support Innovations and Start-ups [X2].
III. Awareness Activities for Promoting Idea Generation and Innovation [X3].
IV. Promotion and Supporting Entrepreneurship Development [X4].
V. Intellectual Property (IP) Generation, Technology Transfer and Commercialization [X5]
VI. Innovative Learning Methods, Courses and Collaborations [X6]
VII. Innovations in Governance and Leadership [X7].

Taguchi Analysis: Innopreneurship versus X1, X2, X3, X4, X5, X6, X7

Table 03: Response Table Signal to Noise Ratios (Larger is better)

Level	X1	X2	X3	X4	X5	X6	X7
1	9.230	10.955	9.757	9.376	9.323	9.300	9.210
2	9.828	8.103	9.301	9.682	9.735	9.758	9.848
Delta	0.599	2.851	0.457	0.305	0.412	0.459	0.637
Rank	3	1	5	7	6	4	2

Table 04: Response Table for Means

Level	X1	X2	X3	X4	X5	X6	X7
1	2.953	3.548	3.120	3.020	2.948	2.948	2.972
2	3.162	2.567	2.995	3.095	3.167	3.167	3.143
Delta	0.208	0.982	0.125	0.075	0.218	0.218	0.172
Rank	4	1	6	7	3	2	5

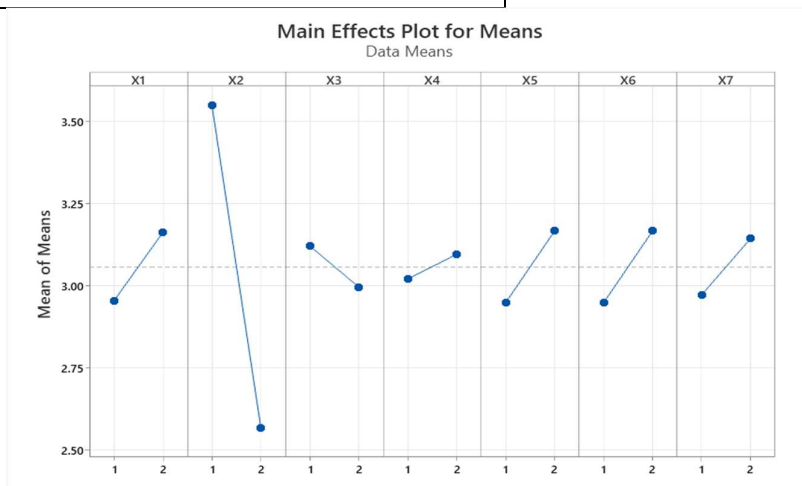


Figure 03: Main effect plot for Means

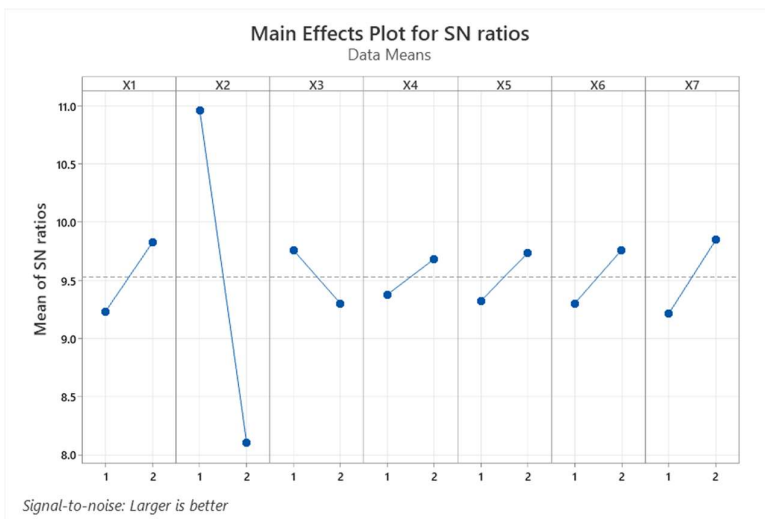


Figure 04: Main effect plot for Means

Figure 03 and 04, explains the variation and ranking level of varioable used for innopreneurship process. The factor ranking as per figure 03 are: X2-X7-X1-X6-X3-X5-X4 whereas from figure 04 the factor are: X2-X6-X5-X1-X7-X3-X4

Conclusion:

The innopreneurship process model in a university setting requires a multifaceted approach that involves fostering a culture of innovation, providing mentorship and support, creating opportunities for funding and investment, and building partnerships with industry. Based on the Taguchi analysis the factor ranking are X2-X7-X1-X6-X3-X5-X4. This also states that Infrastructure and facilities (X2) has highest influence and Promotion and Supporting Entrepreneurship (X4) has lowest influence than other parameters on the Innopreneurship process. Innopreneurship in universities can help to create a pipeline of innovative startups and entrepreneurs who can drive economic growth and create new jobs to the society. This work highlights the entrepreneurial and innopreneurial skills required by the students and thus the organizations for sustainable education for the future. The regression model developed will help the organization to fine tune the innovation and entrepreneurial activities to plan and enhance the innopreneurship process.

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